XGS Energy

Materials science unlocking geothermal energy globally

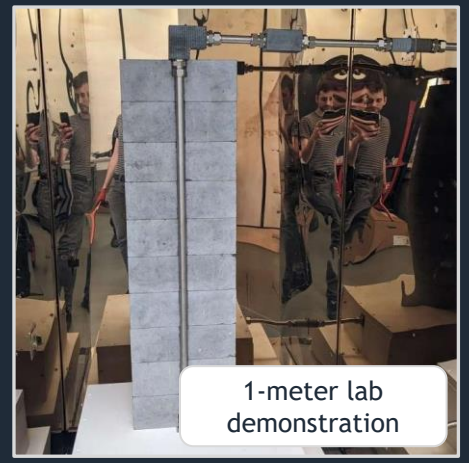
Technology Deep-Dive

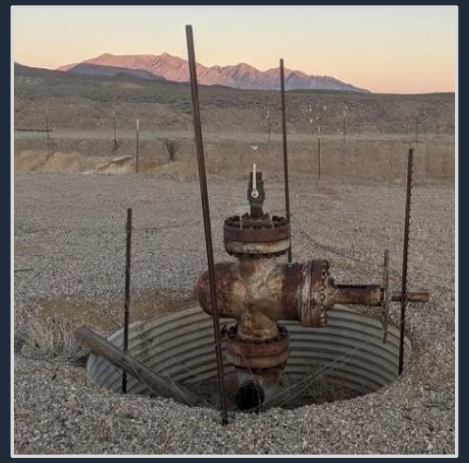
Q1 2025

Roadmap

2021	2022-2023	2024	2025-2026	2026+
Proof of Concept & early testing	Thermal models, Lab build & demos, Shallow demos	Prototype well, brownfield development	First commercial projects, Scale development	Global deployment of utility scale projects

TRE slurry testing under pressure


1-meter lab demonstration



$13.5M Series A (2022)	$20M Series A Ext.	$65M Growth

$400M DevCo + AssetCo

Technology deep-dive summary

	Technology stage	Timing	Milestones
1	Concept & early IP	2021	✓ Evaluate potential impact of TRE approach on geothermal well output ✓ Develop early IP portfolio
2	TRE slurry development	2022	✓ Refine TRE slurry composition for thermal conductivity & scale ✓ Confirm TRE slurry stability and thermal conductivity (target >30 W-m/K; up to 160 W/m-K observed) under subsurface stress condition of >30 MPa
3	Thermal modeling	2022-2023	✓ Evaluate impact of TRE scale and geometry on volume of hot rock accessed, heat flux, and geothermal well output ✓ Develop technoeconomic model to optimize volume of TRE placed vs. cost of energy delivered
4	Lab demonstration	Q1-Q3 2023	✓ Build 1-m lab demonstration from granite blocks and TRE material ✓ Collect empirical evidence of TRE in lab demonstration up to 250°C ✓ Match observed vs. modeled working fluid temperature within 10% error
5	Shallow field demonstration	Q2 2024	✓ Validate TRE settling capacity in “sheath” configuration at 100-m scale ✓ Measure sheath radial thermal conductivity versus sheath porosity
6	“TRE sheath” prototype	Q4 2024 - Q1 2025	• Place >200-m of TRE slurry in “sheath” configuration in prototype well • Circulate working fluid in closed-loop • Demonstrate working fluid temperature >90°C at surface • Match observed vs. modeling working fluid temperature within 10% error
7	Technology roadmap	Ongoing	

Note: TRE = Thermal Reach Enhancement

What is Thermal Reach Enhancement (TRE)?

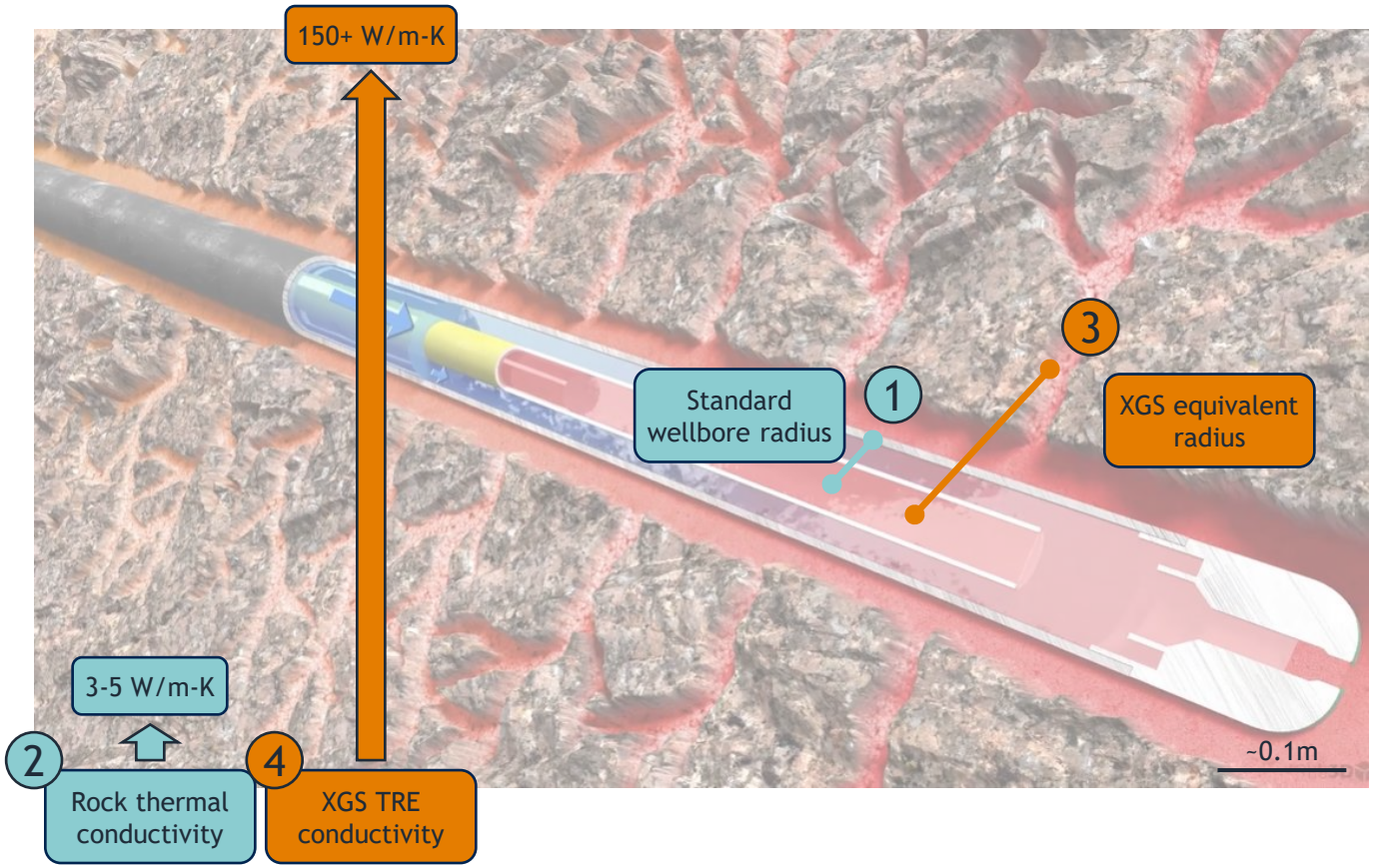
In a standard geothermal well, energy output is fundamentally limited by:

1. Wellbore radius
2. Rock thermal conductivity

XGS breaks this model by introducing thermally conductive material around the wellbore. This creates a new:

3. Equivalent wellbore radius
4. Equivalent thermal conductivity

We call this concept **Thermal Reach Enhancement** – XGS technology increases the "reach" of the wellbore into the hot rock formation, accessing more heat energy without drilling a larger well!



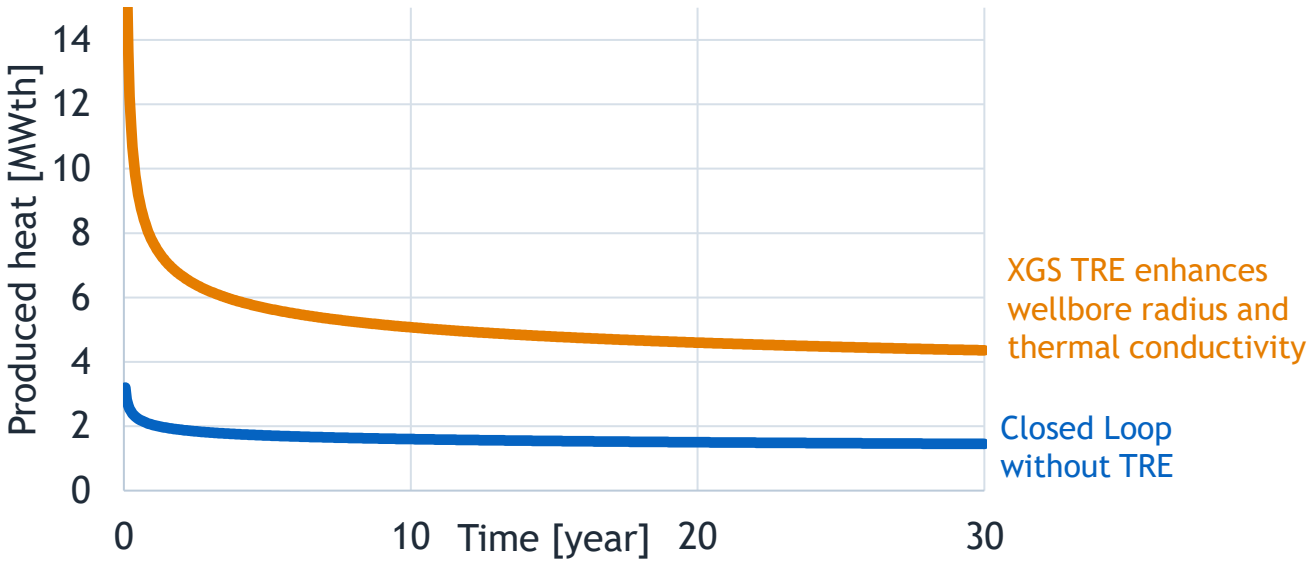
TRE can dramatically impact geothermal energy output

Closed Loop System
Analytical heat transfer formula

$$Q = \frac{4\pi \cdot K \cdot L_w \cdot \Delta T}{\ln\left[\frac{K \cdot t}{\rho \cdot c_p \cdot r_w^2}\right] + 0.80907}$$

Q: Flow rate (m3/sec)
K: Thermal conductivity (W/m.k)
L_w: Harvesting length (m)
ΔT: Temperature drawdown(ºC)
t: Time (sec)
ρ: Resource specific mass (kg/m³)
c_p : Resource specific heat J/(kg.K)
r_w: Hole radius (m)

Illustrative impact of TRE; 3km-300C
Uses simple equation on left; not a real well model



Key variables to maximize thermal power output:
- Increased **resource equivalent thermal conductivity**
- Increased **equivalent wellbore radius**

Note: Analytical formula above is commonly used in oil & gas well testing; see, e.g., Bourdet, D. 2002. Well test analysis: The use of advanced interpretation models.

XGS IP portfolio covers TRE concept & system components

As of Jan 2025

IP Category	Title	XGS ID	Jurisdiction	Status	Updates
TRE Concept, Systems Design & Modeling	Systems and Methods for TRE	0006	US, TW, AU, CA, EP, IN, JP, PH, SA, MX	US, TW, AU, JP allowed	• Clean IPRP received from WIPO 6/24 • US claims 1-16 allowed 1/24, filed claims 17-20 as divisional • TW allowed 04/24, AU allowed 09/24, JP allowed 12/24
	0006 divisional to expand scope	0006-2	US, TW		• US filed 1/24, TW filed 7/24
	GeoHeat Harvesting Enhancement	0007 0007-2	US, TW, AU, CA, EP, IN, JP, PH, SA	US, TW, CA, AU allowed	• Clean IPRP received from WIPO 3/24 • TW allowed 3/24, AU allowed 6/24, CA allowed 6/24, US allowed 10/24
	0007 divisional to expand scope	0007-2	US	US allowed	• US allowed 12/24
	Thermal Reach Enhancement Structures with Improved Energy Transfer	0021	PCT		• Converted to PCT 11/24
TRE Compositions	High Thermal Coefficient Slurry Compositions and Methods Therefor	0005	US, TW, AU, CA, EP, IN, JP, PH, SA, MX	US, TW, AU allowed	• Clean IPRP received from WIPO 6/2024 • TW allowed 11/23, US allowed 11/24, AU allowed 11/24
	0005 divisional to expand scope	0005-2	TW	TW allowed	• TW allowed 12/24
	Compositions and Methods to Form a High Thermal Conductive Sheath	0026	Provisional		• Provisional filed 02/24
	High Thermal Coefficient Grout Compositions and Methods Therefor	0004	US, TW, AU, CA, EP, IN, JP, PH, SA, MX		• Broadens moat by protecting cement compositions that could be alternatives to TRE, though more operationally challenging • Clean IPRP received from WIPO 6/24 • TW examiner comments received; response filed

XGS IP portfolio covers TRE concept & system components

As of Jan 2025

IP Category	Title	XGS ID	Jurisdiction	Status	Updates
TRE Placement (operational methods & tools)	TRE Flowback Prevention Compositions And Methods	0010	PCT, TW	TW allowed	• ISR/WO and shows claims with clean search; arguments prepared for inventive step for remaining claims • TW allowed 8/24
	Dual Mode Fracture Tool And Methods Therefor	0019	PCT		• Converted to PCT 10/24
	Improved Systems & Methods to Form High Thermal Conductive Sheath	0027	Provisional		• Provisional filed 02/24
	Systems and Methods to Place a Thermally Conductive Sheath in a Geothermal Well	0028 0028-2	Provisional		• Provisional filed 4/24 • Provisional with more detailed claims filed 12/24

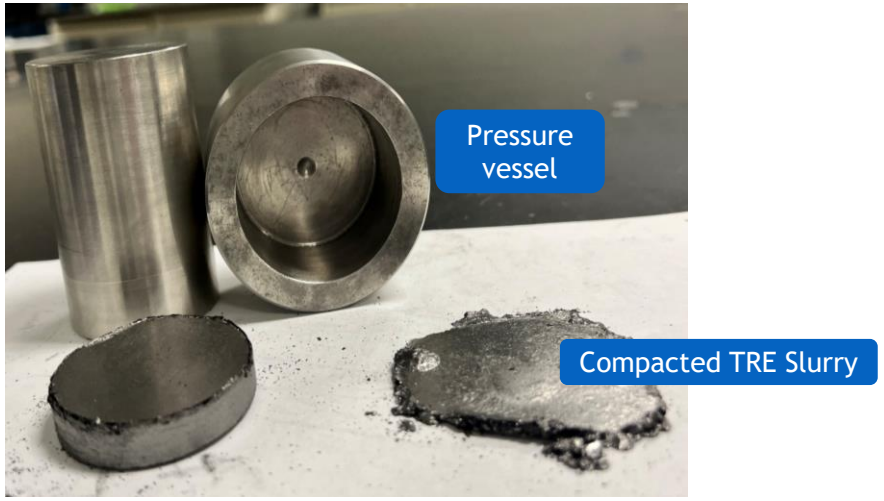
IP Category	Title	XGS ID	Jurisdiction	Status	Updates
Other (Non-core / legacy IP)	Geothermal Heat Harvesters	0022	US, AU, CA, JP, EP	US AU CA JP EP allowed	• Europe allowed & registered as unitary patent 3/24
	INSULATED WELDED JOINT FOR PIPE-IN-PIPE SYSTEMS	0003	US, EP, JP	JP allowed	• Clean IPRP received 11/23 • JP allowed 12/24

How does TRE slurry work?

Extensive testing conducted to optimize thermal conductivity, viscosity, pumpability, and scale

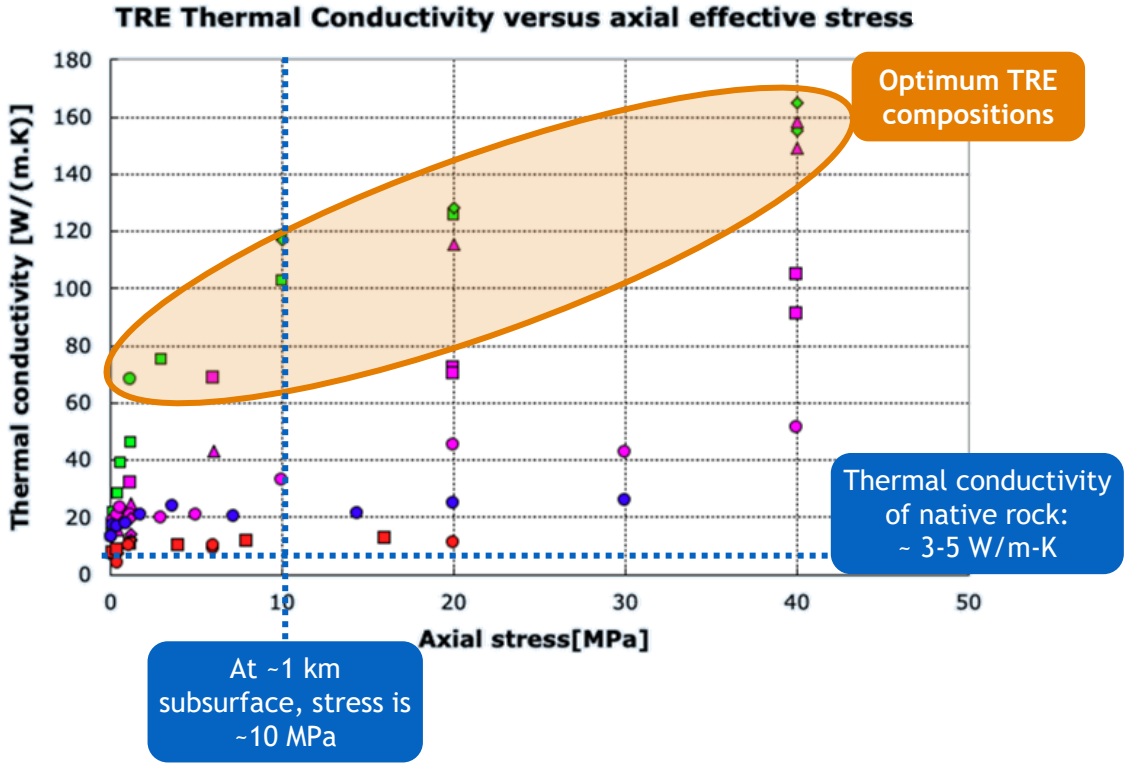
How demonstration works:

- TRE slurry compositions are compacted under same magnitude of pressure experienced subsurface
- Thermal conductivity measured on compacted material



The results:

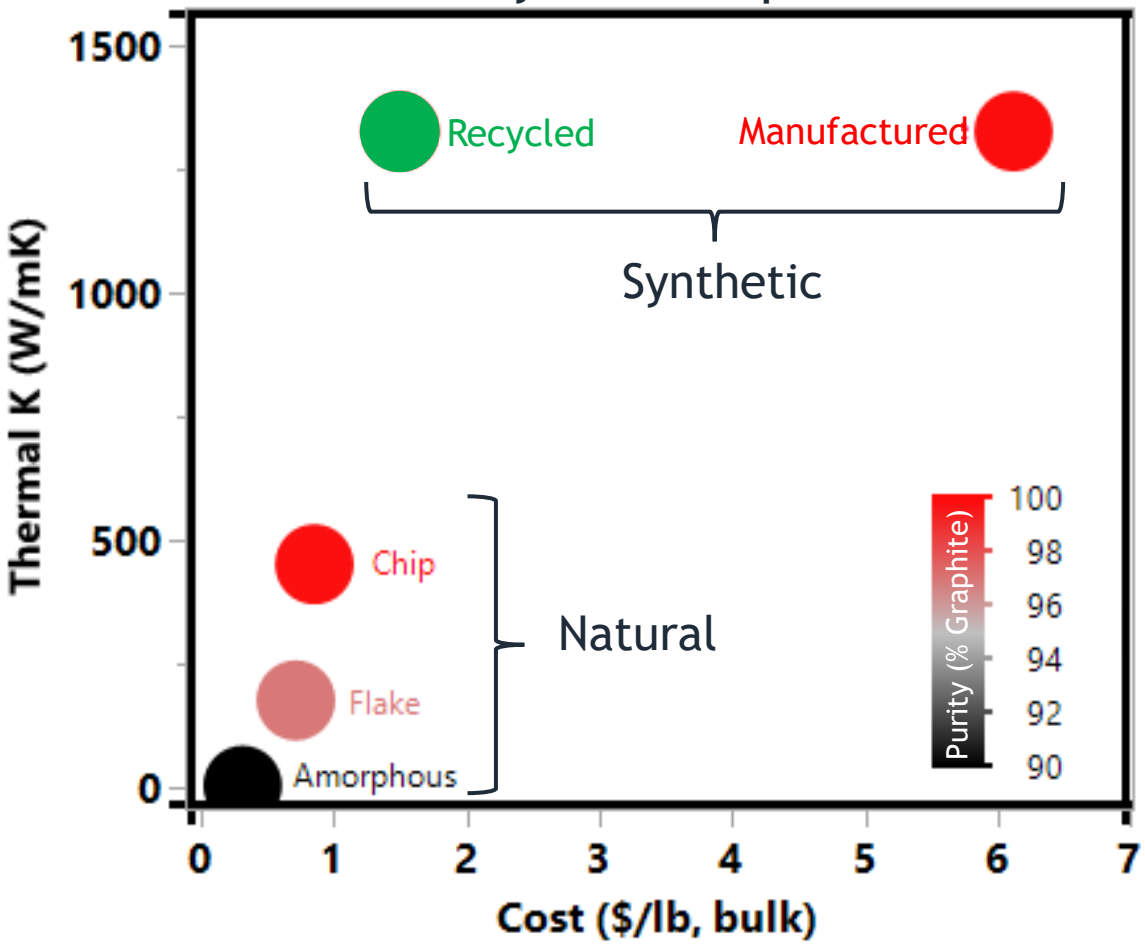
- Optimum TRE slurry composition shows thermal conductivity 50X that of native rock



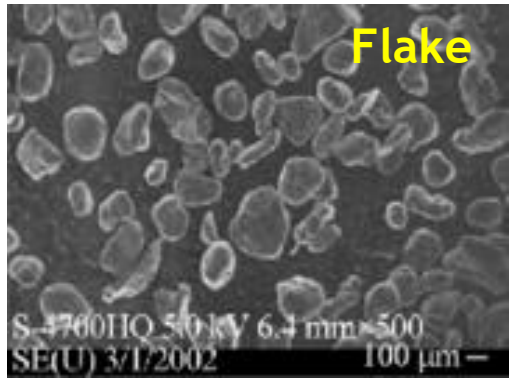
Carbon-based TRE material properties and costs vary, providing abundant space for further optimization

Illustrative view for one TRE slurry component: Graphite

Natural and Synthetic Graphite Products

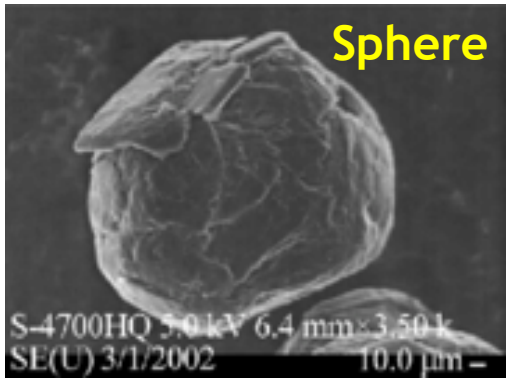


Graphite Morphologies



Pros:
- Abundant and lower cost
- Slurry: Better suspension
- TRE: Good compressibility

Cons:
- Slurry: Complex rheology
- Placement: Prone to flowback



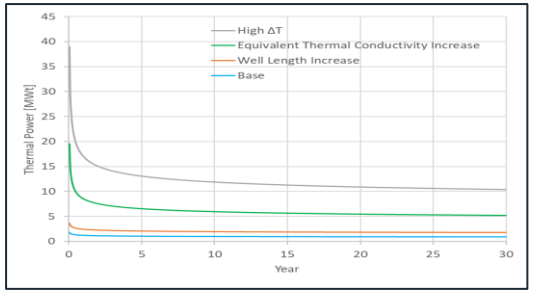
Pros:
- Slurry: Known rheology
- Placement: Permeable for fluid flowback
- TRE: Good compressibility

Cons:
- Custom and higher cost

Source: Zhang et al., *Environ. Sci. Technol.* 2023; Trimmer et al., *Electronics Cooling* 2020; Tian et al.; *J. Hazard. Mater.* 2023; Jara et al., *Int. J. Min. Sci. Technol.* 2019.

XGS Hybrid Thermal and Power Modeling Approach

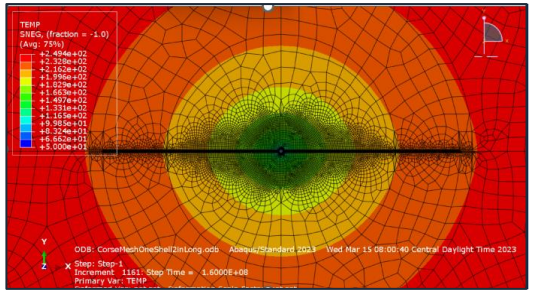
Analytical Model



Advantages
- Provide fast, stable solution

Limitations
- Restrict to special type of problems

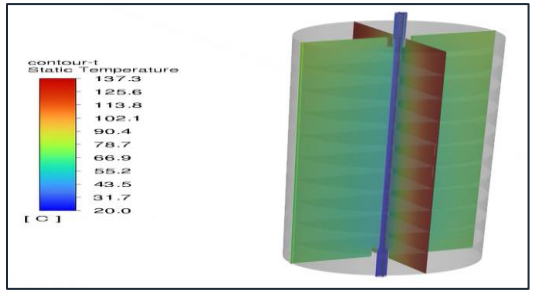
Finite Element Method



Advantages
- Physics Based model
- Modeling of complex geometries
- Different types of material properties
- Heat Transfer prediction

Limitations
- Computationally intensive
- FEM simulations must be validated against physical data

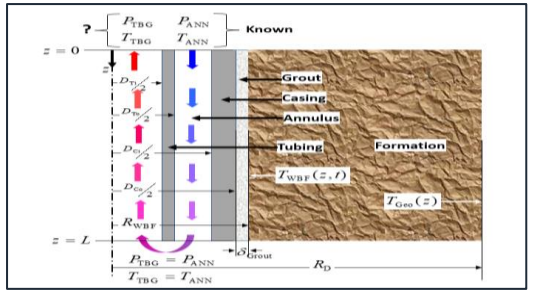
Computational Fluid Dynamics



Advantages
- Physics Based model
- Modeling of complex geometries
- Different types of material properties
- Fluid & Heat Transfer prediction

Limitations
- Computationally intensive
- CFD simulations must be validated against physical data

Coupled Wellbore Model



Advantages
- Provide fast, stable solution
- User friendly interface
- Tool for well design and engineering optimization

Limitations
- Limited to special type of problems and requires development of new configurations

TRE Formation Thermal Model

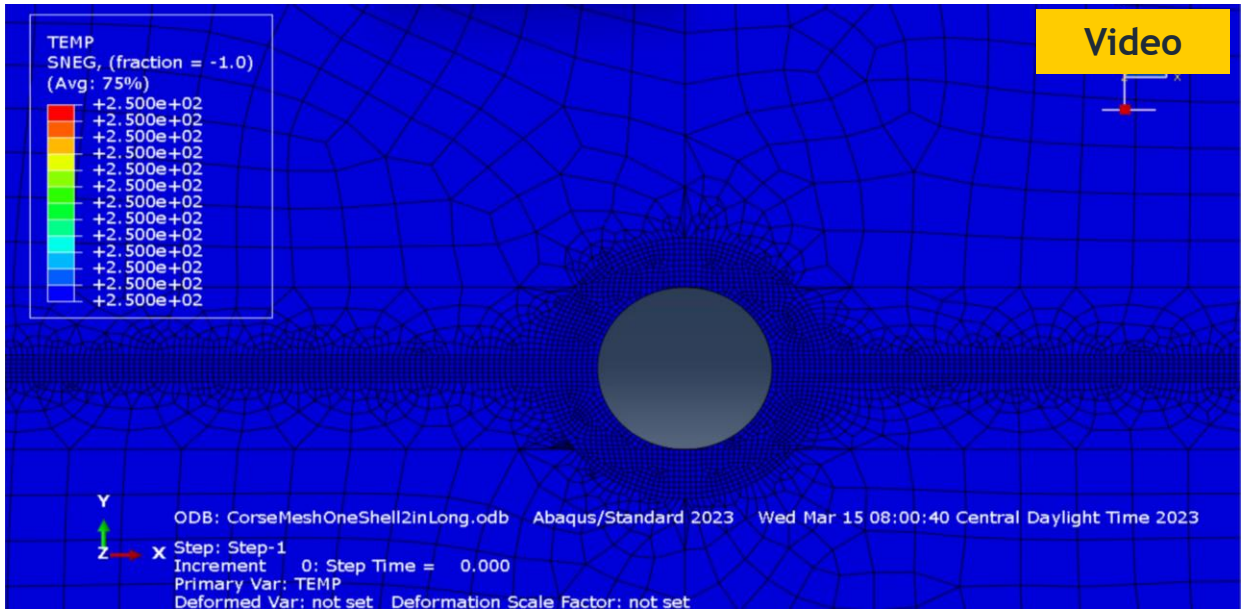
Thermal Power Output

Note: TRE = Thermal Reach Enhancement

Finite Element Model

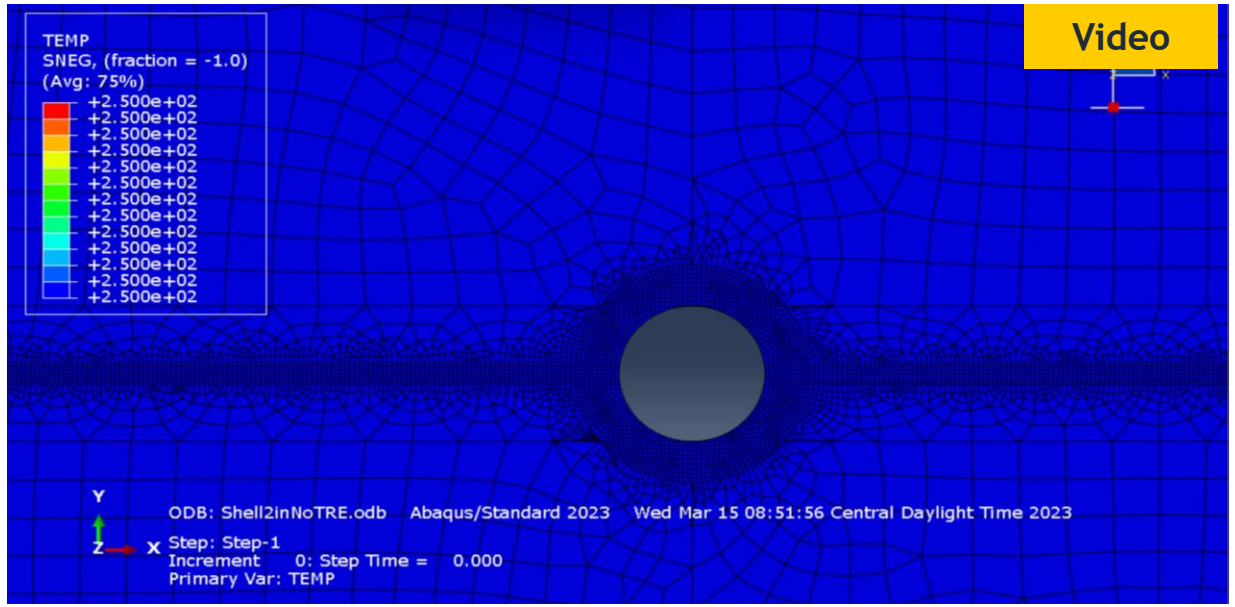
Transient Heat Transfer Analysis

Case with TRE



- Rock temperature: 250°C
- Step 1 – transient 5 year – wellbore temperature 50°C

Case without TRE



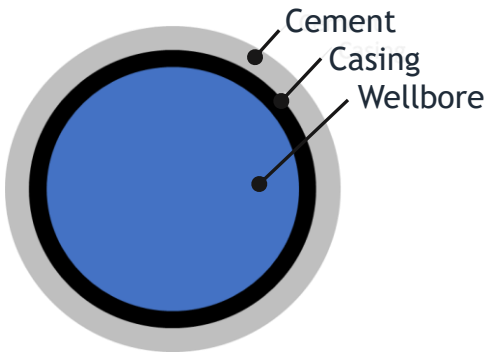
- Rock temperature: 250°C
- Step 1 – transient 5 year – wellbore temperature 50°C

Source: XGS Heat Transfer model; Finite Element Analysis (FEA) with Abaqus

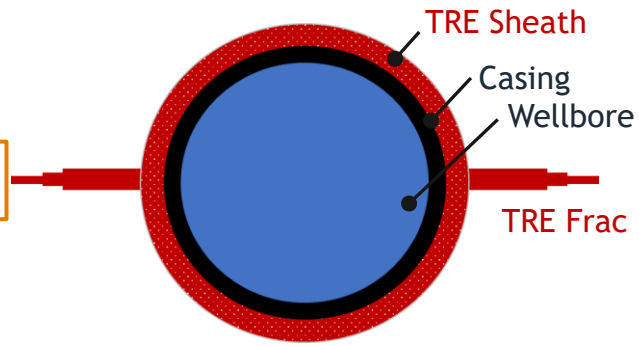
Modeling demonstrates ability of TRE to access 30%+ more hot rock and increase heat flux 300%

Top-down view of wellbore

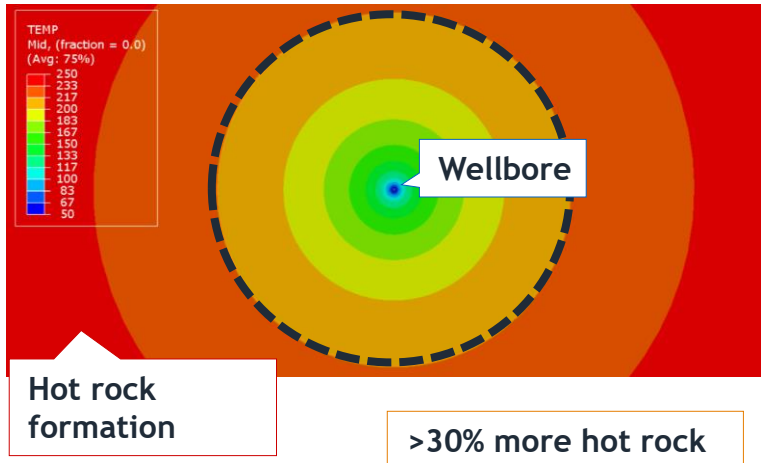
No TRE

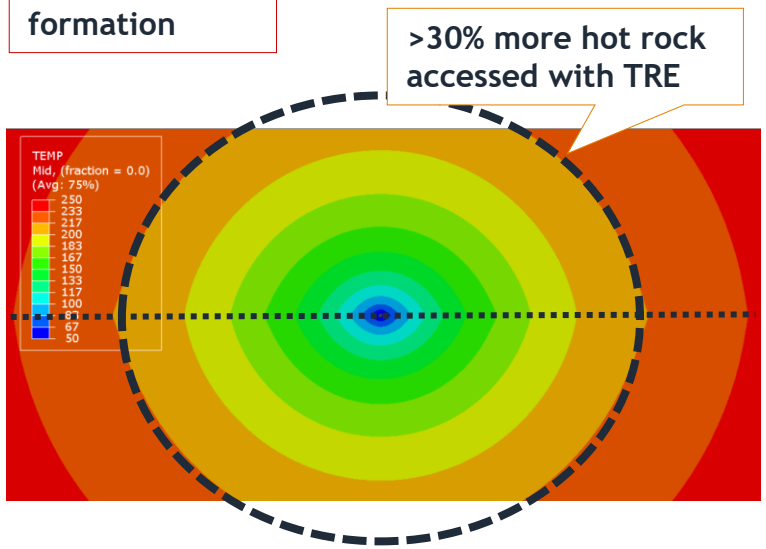


TRE

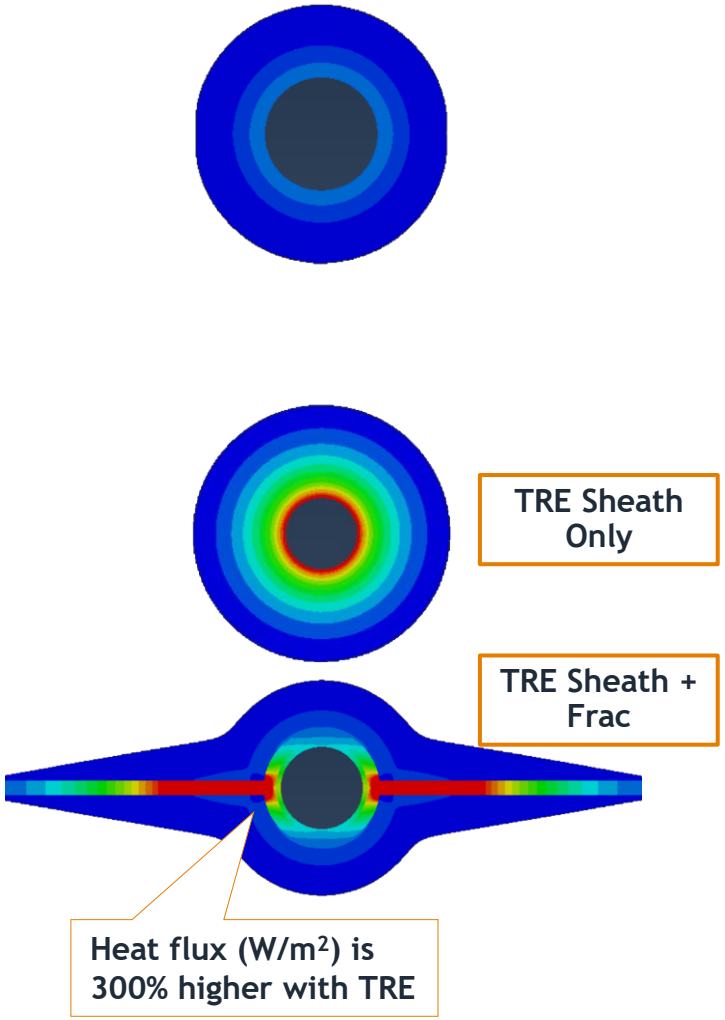


Temperature map of wellbore





Heat Flux at wellbore

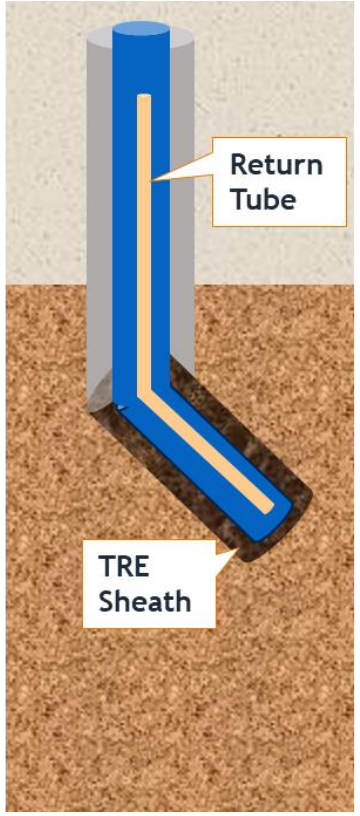


Note: TRE = Thermal Reach Enhancement; Source: XGS Heat Transfer model; Finite Element Analysis (FEA) with Abaqus

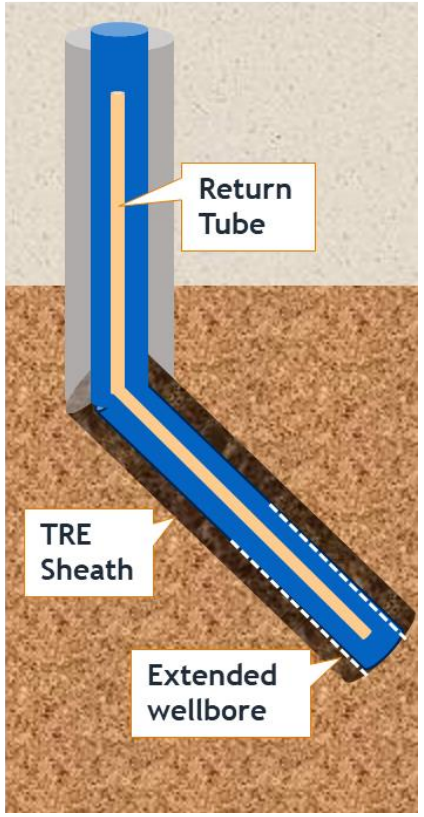
Thermal modeling used to evaluated different TRE geometries

Illustrative

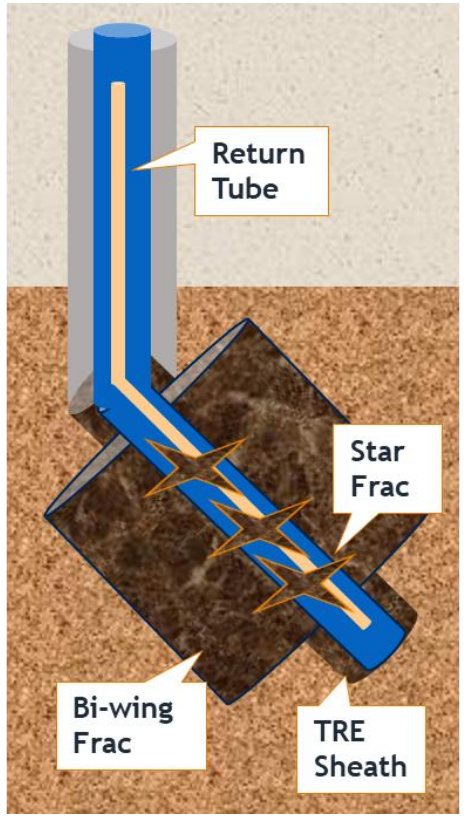
Cased Hole TRE Sheath – No well extension



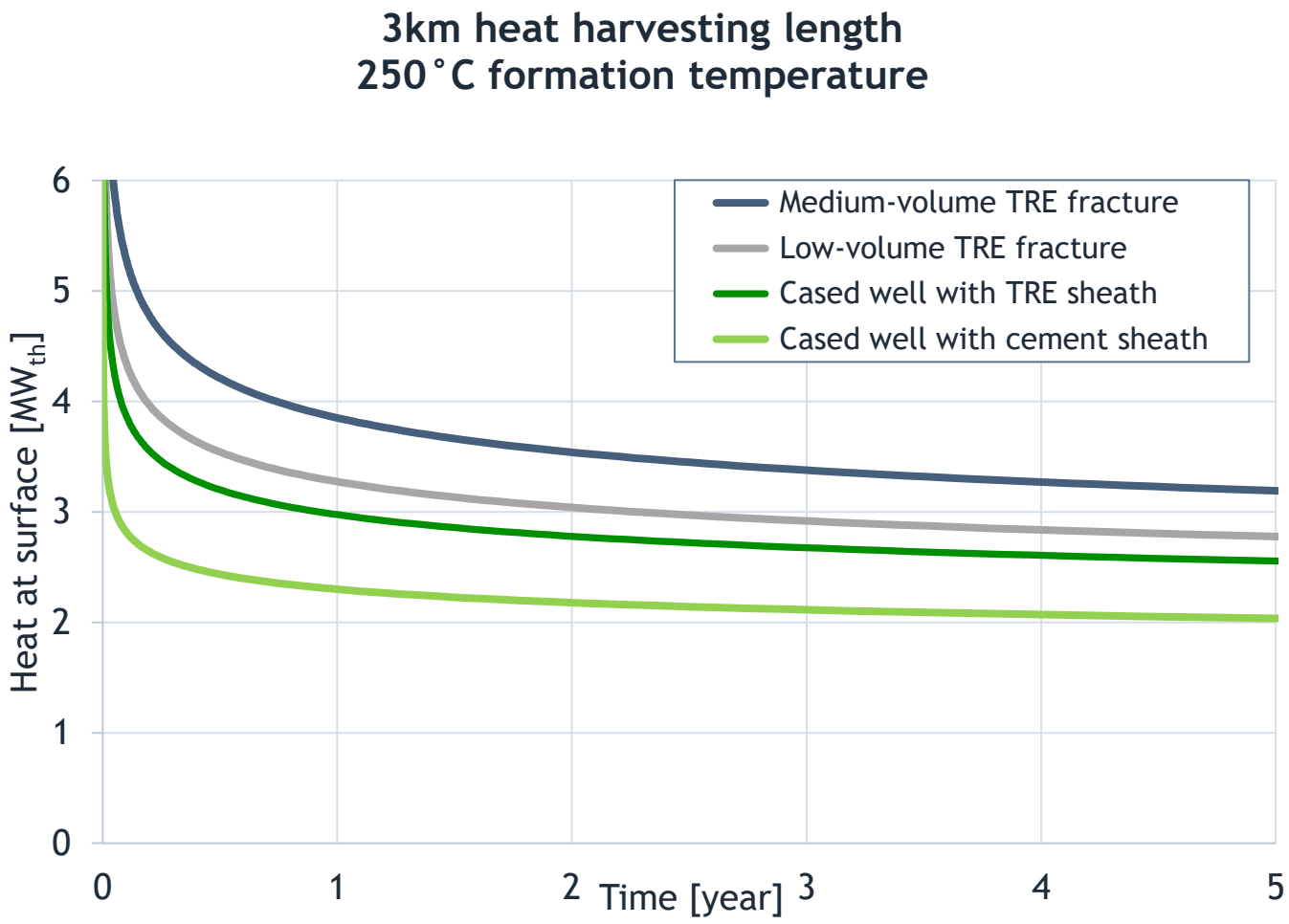
Cased Hole TRE Sheath – Well extension or new well

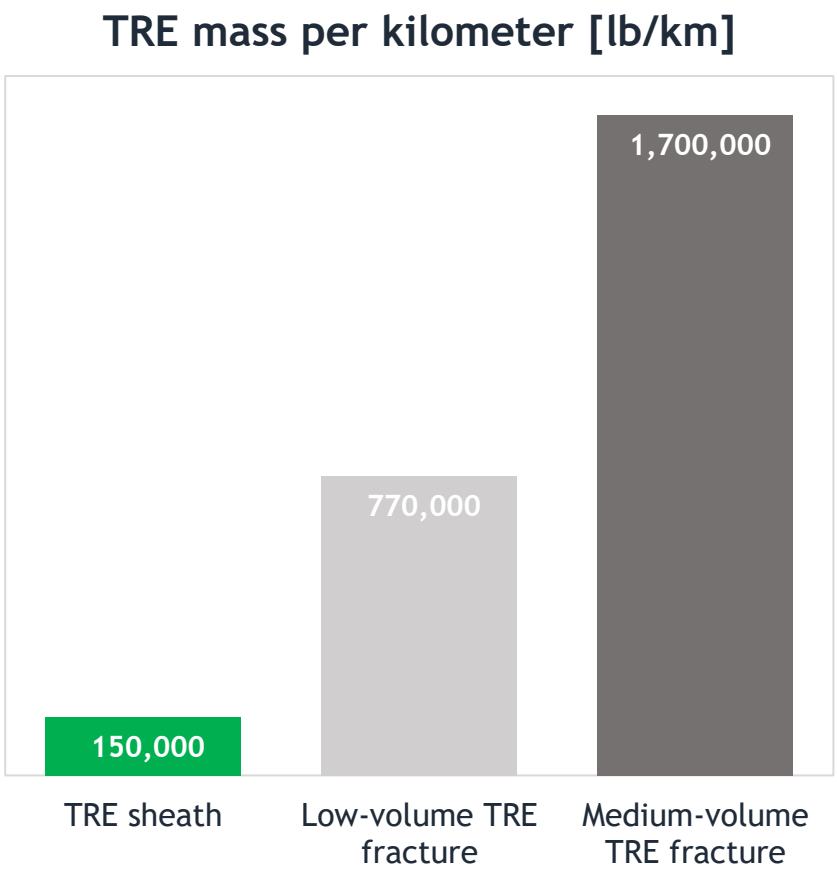


TRE Star + Bi-wing



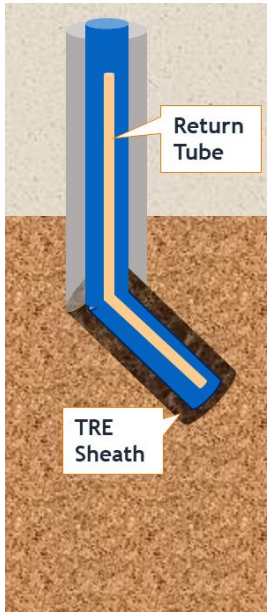
Modeling demonstrates enhancement of thermal energy output with different TRE configurations

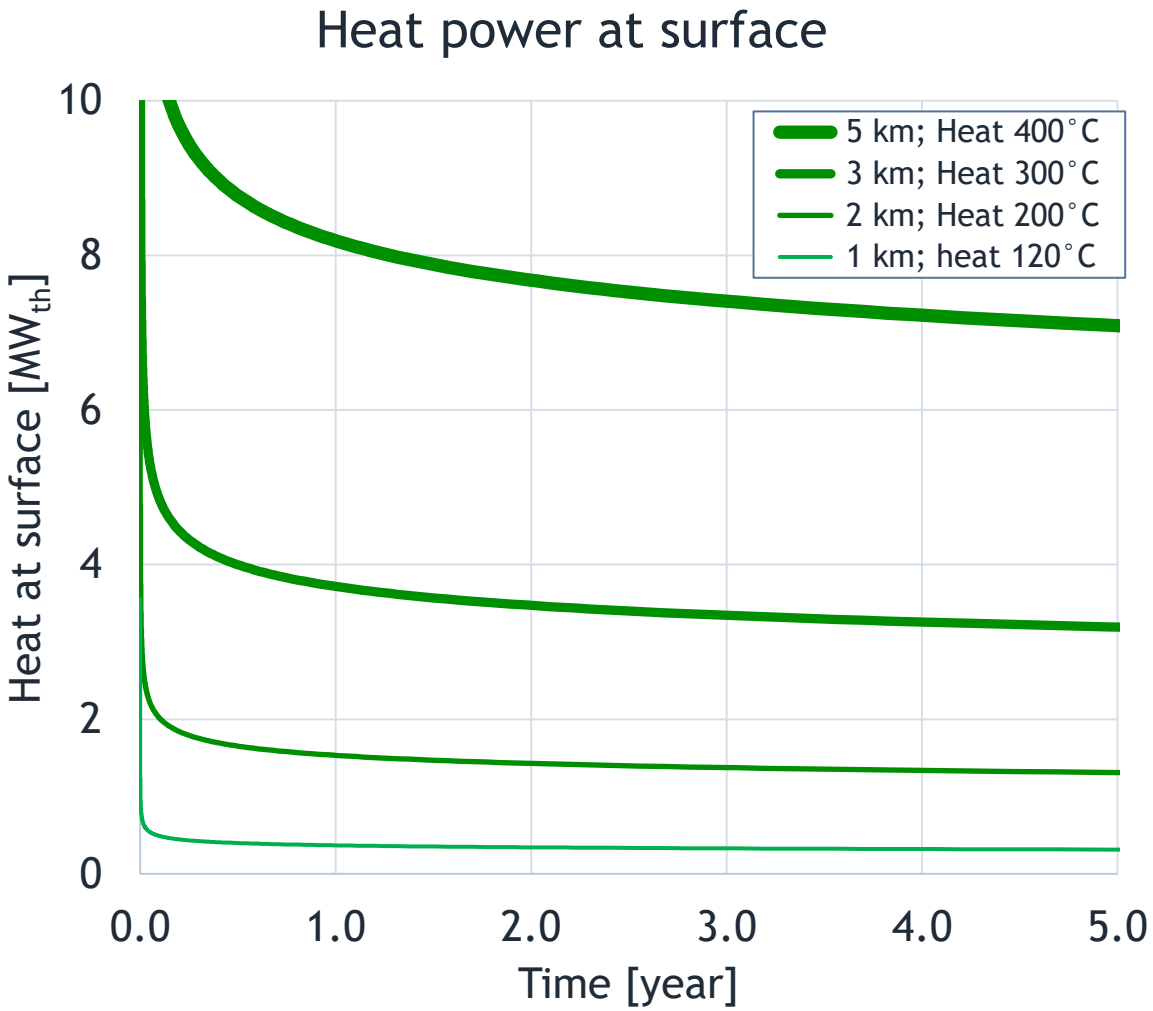


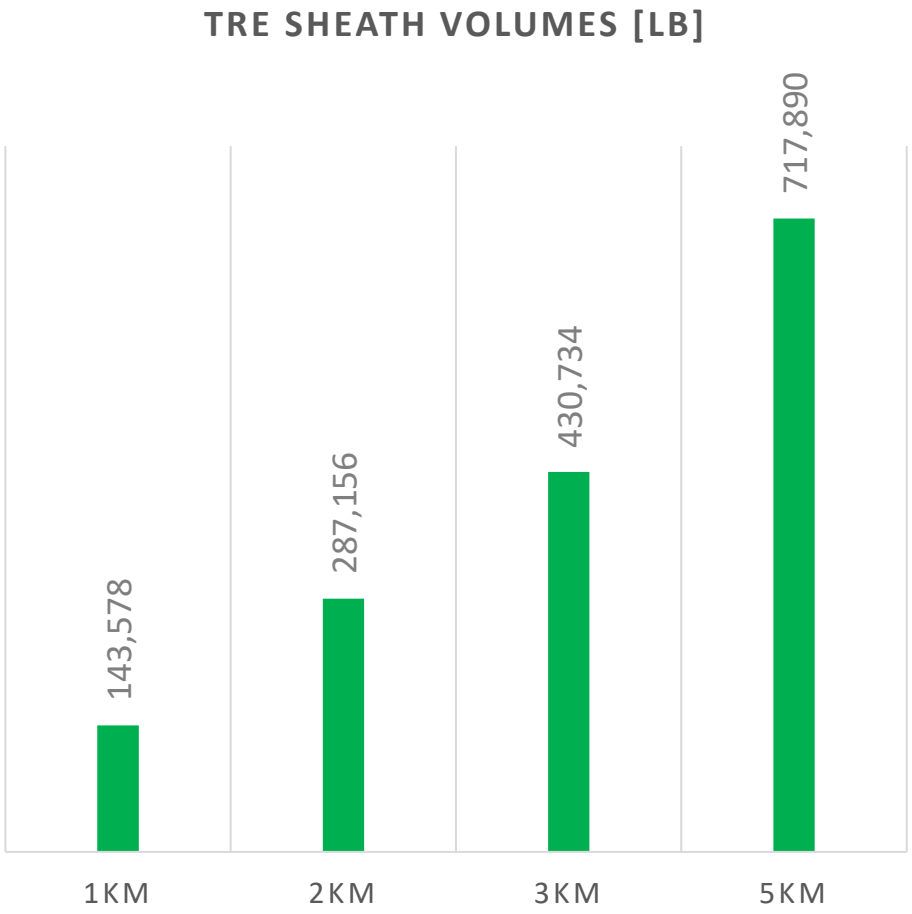


Source: XGS Hybrid Thermal and Power Model

TRE sheath: Impact of higher temperature resource





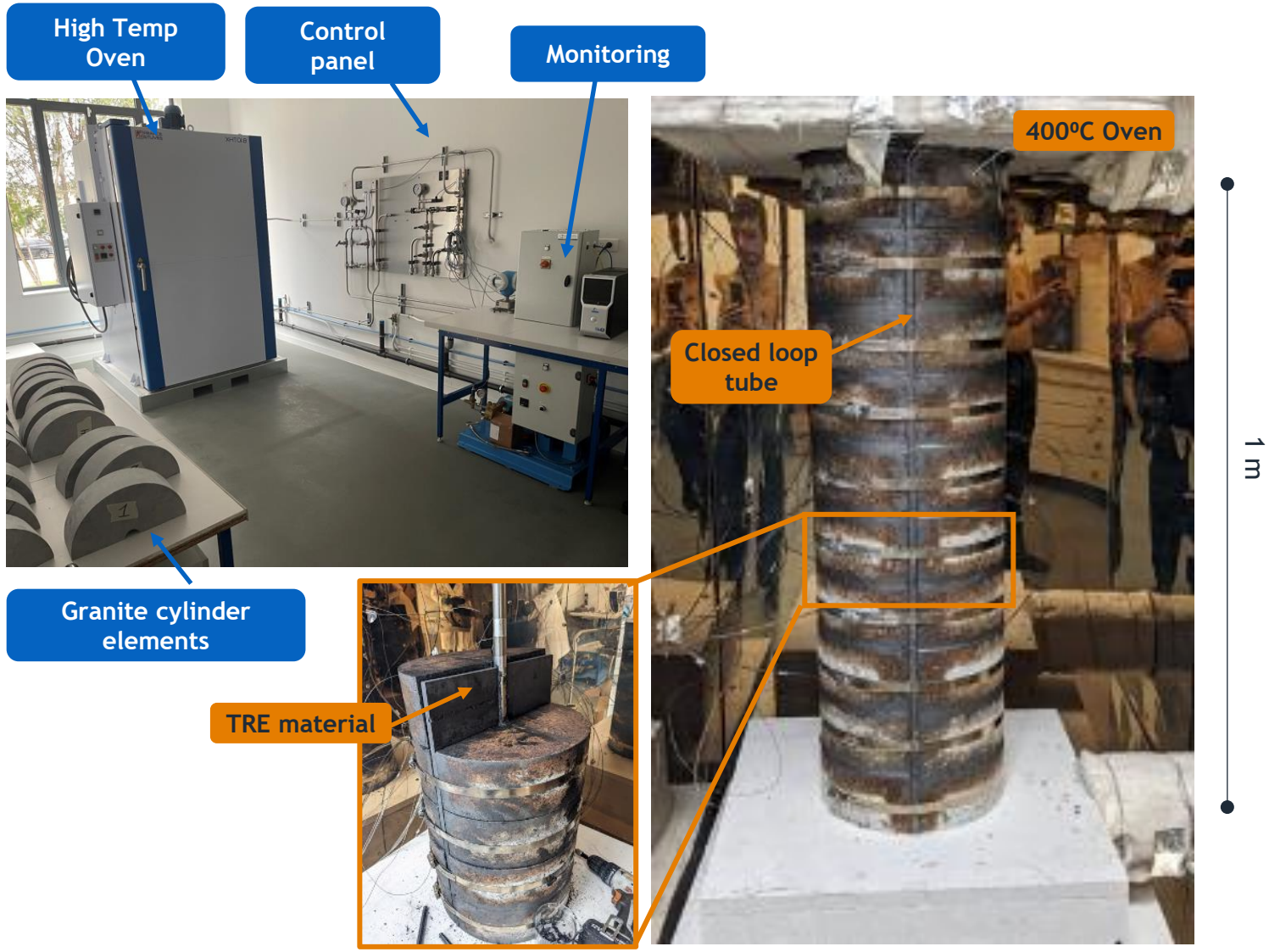


Source: XGS Hybrid Thermal and Power Model

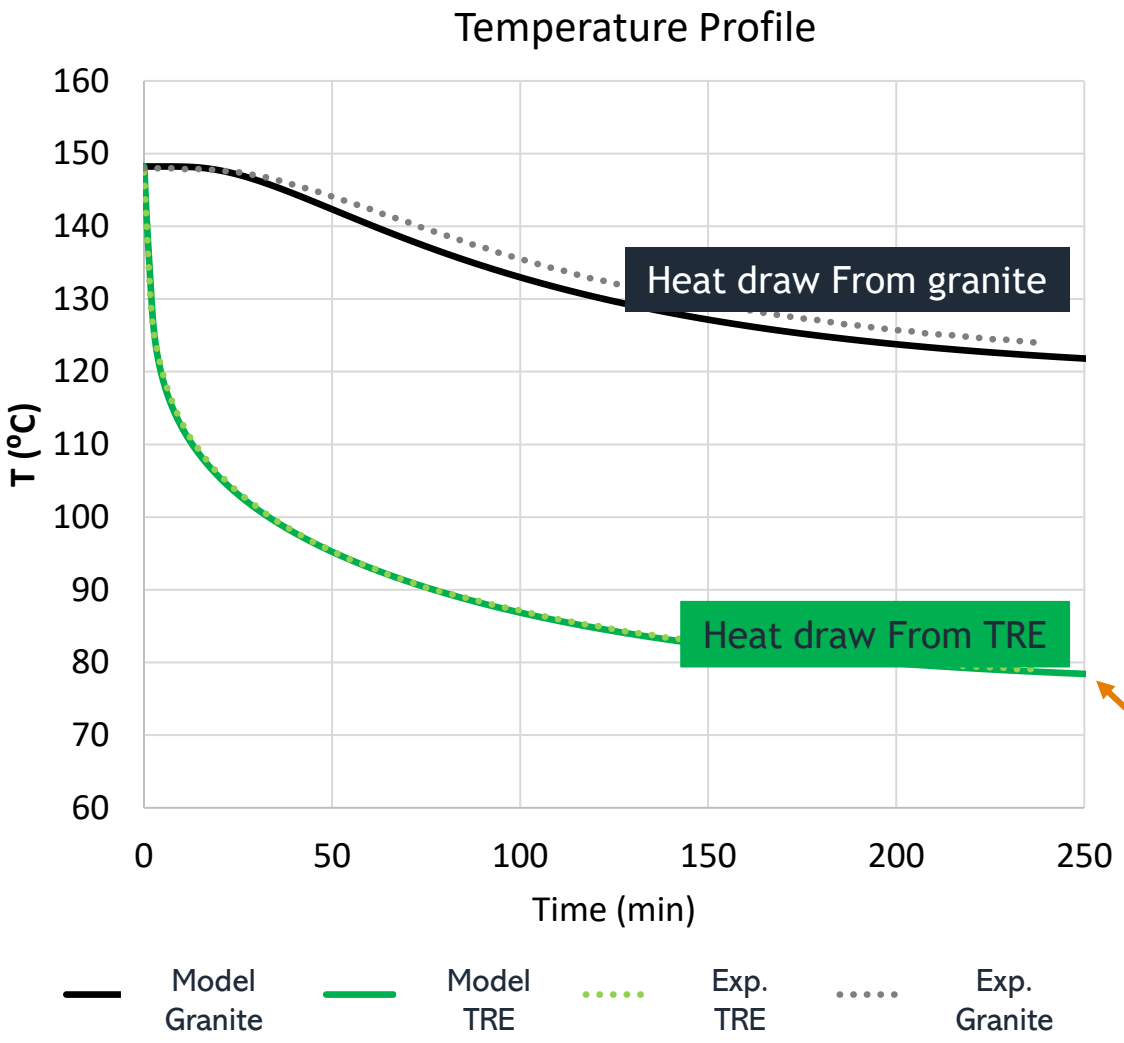
High Temperature Lab Demos: 1 Meter Scale

How demonstration works:

- Pack a meter-scale "column" composed of large granite blocks and TRE material around a central water pipe
- Column is built within an oven with max temperature of 400°C
- Outlet water temperature reflects the magnitude & efficiency of heat harvesting



First-of-kind lab demonstrated empirical evidence to match TRE model



The results:

- Good match between experimental results and numerical model
- TRE shown to drawn 50% more heat than granite alone
- XGS proprietary model refined to replicate/predict experiment results within 2%. Same model is used to evaluate heat energy potential of commercial well
- Heat recovery increased up to 90% for the case with TRE vs no TRE

Temperature (ºC)	TRE thickness (mm)	TRE vs. No-TRE Case Heat Recovery Increase (%)
150	10	45%
250	10	60%
150	20	60%
250	20	90%

Experiment Source: XGS High-Temperature Lab; Model Source: XGS Hybrid Thermal and Power Model

100-m Field Test Demonstrated Ability to Mix, Pump, and Place TRE at Commercial Scale

100-m shallow well demonstration validated ability to scale TRE system for commercial operations

Shallow well results	Commercial impact
Mixed commercial-size batch of slurry formulation with 20 vol% solids content	✓ On track to optimize solid content to 30 vol% target to reduce rig days required for TRE placement and minimize chemical costs
TRE slurry formulation had pumpability of <10 Bc	✓ Achieved commercial pumpability target of <50 Bc, which is required for use of commercially available pumping equipment
Pumped TRE at rate of 0.2 bpm to fill well annulus (for PVC pipe with 1” pumping tube diameter)	✓ On track to achieve target pumping rate of 1-3 bpm in commercial well; pumping rate scales proportionally pressure rating of well materials (steel for commercial) and pumping tube diameter (3” diameter for commercial)
TRE settled at rate of 5.5 ft/min	✓ Achieved commercial settling target of >1 ft/min, reducing rig time and costs
TRE settled with porosity of 68%, optimizing thermal conductivity	✓ Achieved porosity of <70%, which is required to realize commercial target thermal conductivity of >10 W/m-K

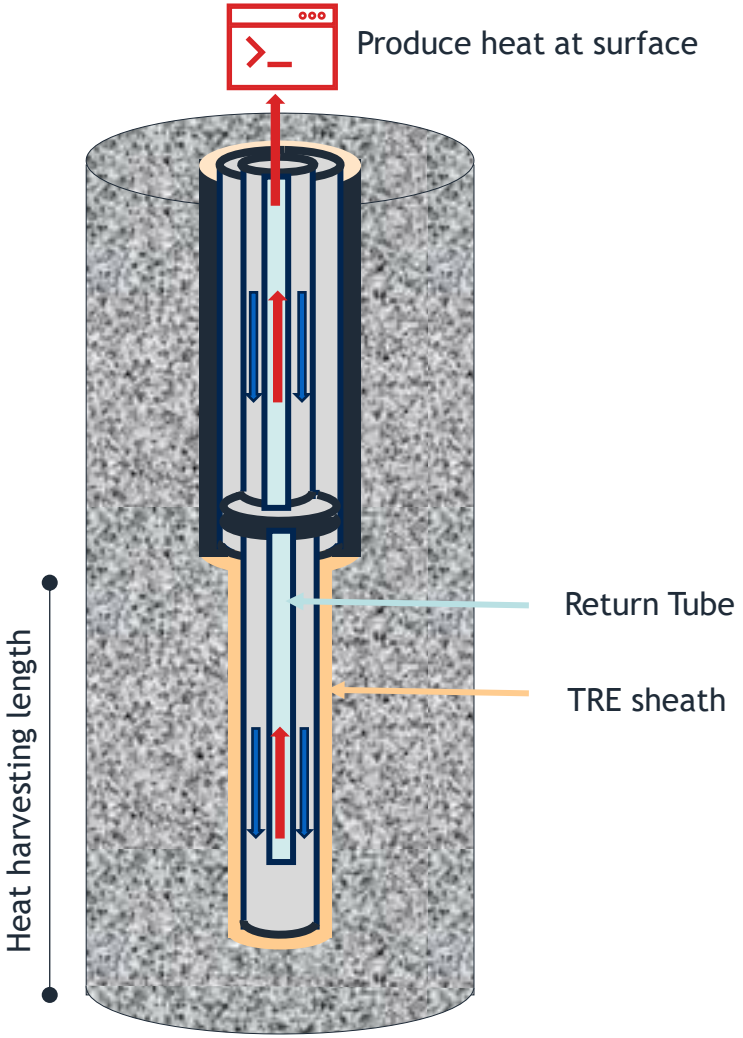
Ongoing: TRE sheath prototype advancing XGS technology to full commercial readiness

Objectives:

1. Enter an existing well
2. RIH liner and place TRE sheath
3. RIH insulated tubing
4. Produce heat at surface

Success criteria:

1. Build operational skills: Enter the well, clean it, RIH liner, place TRE sheath, RIH insulated tubing
2. Achieve homogenous TRE sheath placement along casing
3. Achieve minimum TRE sheath thermal conductivity of 10 W/m-K
4. Test range of working fluid flow rates
5. Achieve output temperature >90°C (dependent on resource temperature)
6. Match observed vs. modeled working fluid temperature with variable flow rate



Note: RIH = Run In Hole

Prototype site selection process - evaluation criteria

Prototype site evaluation criteria		Target condition	Comments
Thermal output	Resource temperature [°C]	>200	The higher the resource temperature, the larger the changes in working fluid temperatures will be, enabling faster, simpler data analysis. Tradeoff is above 250°C may need to cool wellbore to install return tube and use off-shelf monitoring equipment
	Harv. length [km]	>0.5	The longer the harvester length, the larger the changes in working fluid temperatures will be, enabling faster, simpler data analysis. Tradeoff is increasing harvester length increases prototype cost.
	Well diameter [in]	12¼	Smaller diameter wells are incompatible with most common well liner and return tube products. Use of a 12¼-diameter well enables fastest procurement of prototype components to complete as soon as possible
Operations	Well condition	Only cleaning required	Well cleaning is acceptable and likely required at most prototype sites. Sites with reports indicating well instability will not be selected.
TRE sheath placement	Lost zones	Minimal or none	If encountered, lost zones will need to be plugged at added cost.
	Well deviation [°]	<45	TRE sheath placement uses a settling process that is faster the closer the well is to fully vertical (i.e., 0° deviation)
	Frac gradient [sg]	>1.2	The TRE slurry will need to be designed for wellbore strengthening if the frac gradient is below 1.2 sg

Note: sg = fluid weight equivalent of fracture pressure (the pressure at which formation/well integrity would be compromised, creating fractures).

Tech roadmap: XGS aggressively advancing system optimization

Pre-2022 & 2022

Modeling & simulation

Technology development

Commercial operations

Conceptual design and establishment of IP portfolio

TRE slurry development
- Slurry composition
- Rheology
- Particle size & morphology
- Mechanical & temperature durability
- Compaction & pressure testing
- Loading approach design

Series A

2023

Proprietary thermal modeling
- Full system modeling (heat transfer, wellbore, thermal power output)
- Validation with commercial software

Senior technical & commercial hires

Lab demonstration, 1-m scale
- Thermal modeling validation in first-of-kind lab (m-scale, up to 350°C)
- Flow rate optimization
- TRE thickness optimization

Shallow field demonstration, 100-m scale
- TRE slurry materials & loading optimization (large-scale mixing, downhole pump test)
- TRE placement testing (suspension and compaction test)

2024

Series A Ext.

Proprietary thermal modeling
- Integration of subsurface geology database
- Further development of proprietary subsurface resource evaluation workflow

Prototype
- Full system execution in operational environment
- Installation of km-scale heat harvesting well
- Flow rate optimization
- Surface heat measurement

Further TRE engineering
- Dedicated fluids lab
- Next-gen TRE design
- Next-gen TRE formulation shallow field testing

Today

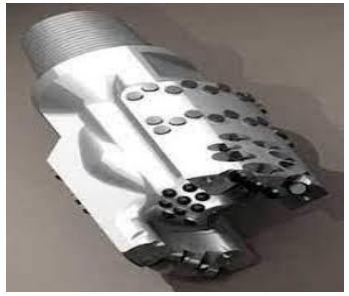
2025+

Commercial projects
- Project(s) in target markets at 5+ MW scale
- Operation at >300°C resource temperatures

Commercial capability build
- US development team (origination, permitting & regulatory, leases)
- Subsurface exploration team(s) by focus geography (geologist, geophysicist, geomechanics engineer, reservoir engineer)
- Procurement & operations

Continuous system development
- Surface power plant working fluid optimization
- Integration of latest drilling technology advances
- Rig customization for XGS tech
- Technology deployment in range of subsurface conditions (wet/dry, corrosive, etc.)

XGS to benefit from continuous industry advancement enabling faster, higher-temperature drilling



Selection of recent Oil & Gas and Geothermal industry innovations that can be leveraged in XGS wells:

- **Metal to Metal power section**, as a replacement for the rubber power sections in the industry.
- **Particle Impact Drilling (PID)**: Conventional drill bits crush and grind rock, but PID does not rely on weight-on bit and torque to mechanically break the rock.
- **Drill bit, PDC cutter** diamond layers improves the cutters' thermal and abrasion resistance
- **Plasmabit** tool, reduces vibrations and improves stability when drilling with rotary systems in the hard and abrasive formations commonly encountered in deep and hot geothermal projects.
- **Gyrotron-powered drilling** platform vaporizes boreholes through rock and provides access to deep geothermal heat without complex downhole equipment.

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